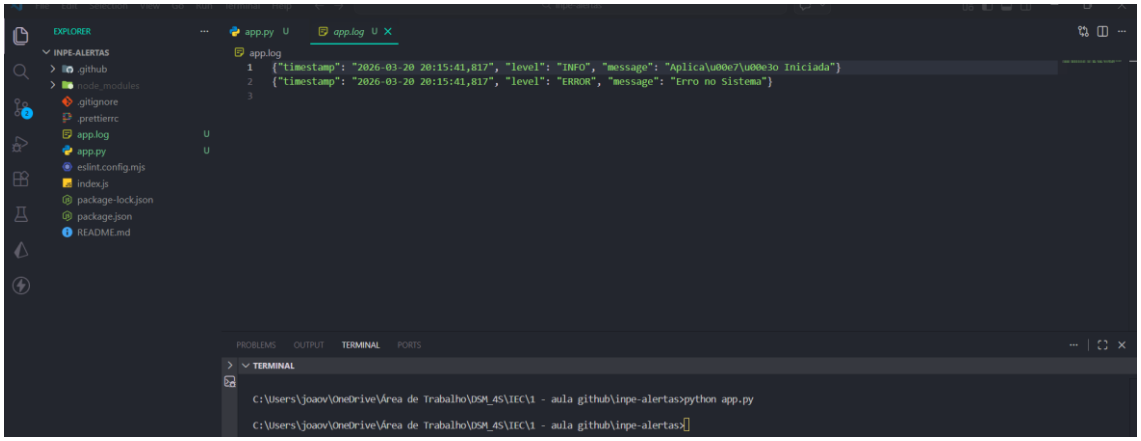
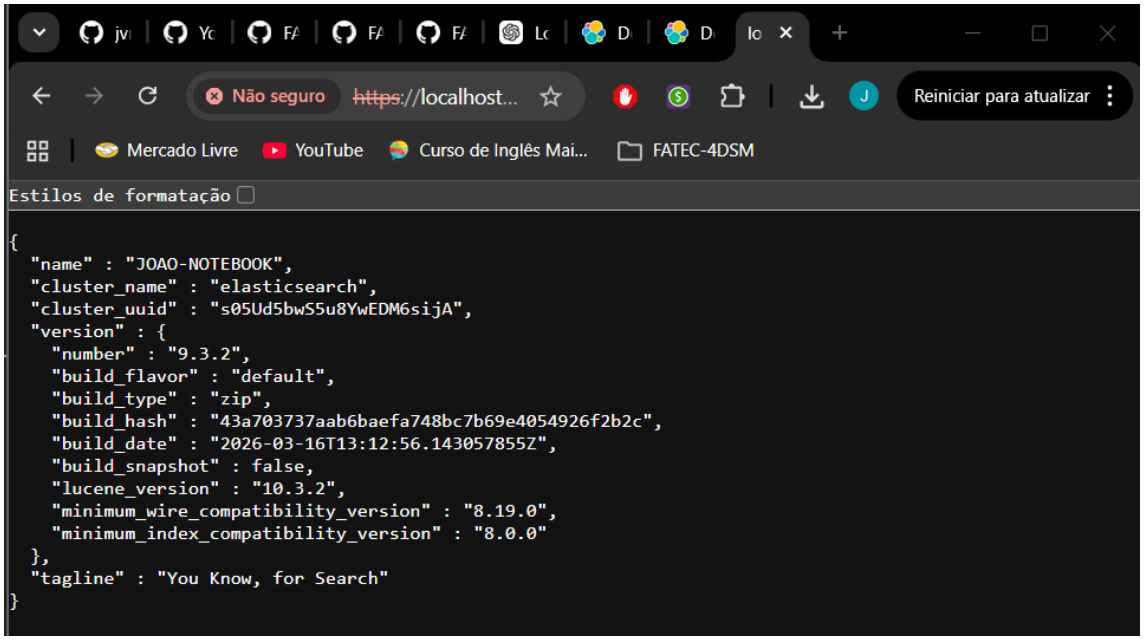
Criando o app.py

Rodando o app.py para gerar log



Validação do funcionamento do Elasticsearch via endpoint local, retornando informações do cluster em JSON.

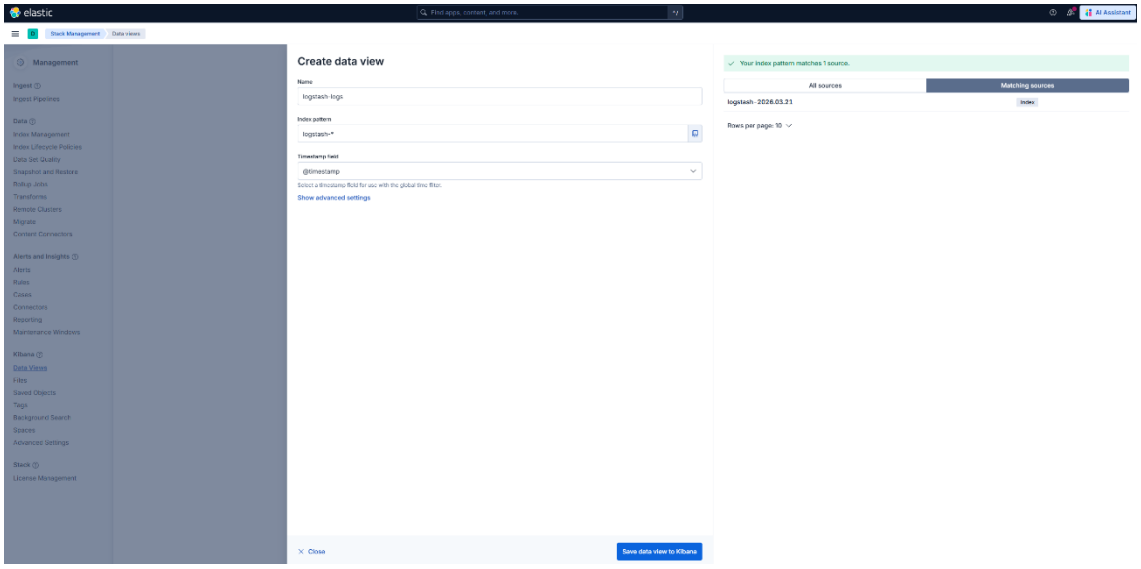
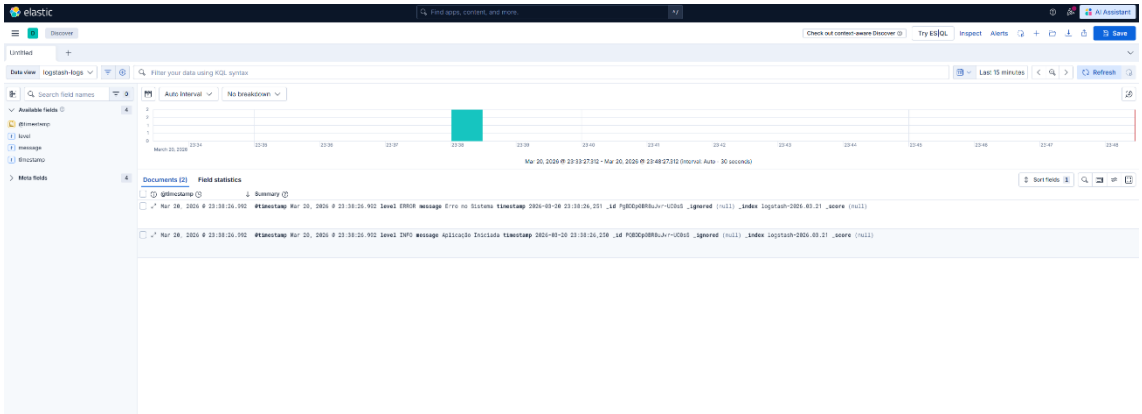


Acesso ao Kibana e escolha da exploração manual do ambiente.

Execução do Fluentd para coleta e envio de logs ao Elasticsearch.

```
C:\Users\joaov\OneDrive\Área de Trabalho\DSM_4S\IEC\1 - aula github\inpe-alertas>fluentd.bat -c fluent.conf
2026-03-20 23:37:02 -0300 [info]: init supervisor logger path=nil rotate_age=nil rotate_size=nil
2026-03-20 23:37:02 -0300 [info]: parsing config file is succeeded path="fluent.conf"
2026-03-20 23:37:02 -0300 [info]: inaccessible include directory was specified path="/etc/fluent/conf.d"
2026-03-20 23:37:02 -0300 [info]: gem 'fluent-plugin-elasticsearch' version '6.0.0'
2026-03-20 23:37:02 -0300 [info]: gem 'fluentd' version '1.19.2'
2026-03-20 23:37:03 -0300 [info]: Oj is not installed, and failing back to JSON for json parser
2026-03-20 23:37:03 -0300 [info]: using configuration file: <ROOT>
  <source>
    @type tail
    path "./app.log"
    pos_file "./fluentd.pos"
    tag "app.logs"
    format json
    <parse>
      @type json
      unmatched_lines
    </parse>
  </source>
  <match app.logs>
    @type elasticsearch
    host "localhost"
    port 9200
    logstash_format true
  </match>
</ROOT>
2026-03-20 23:37:03 -0300 [info]: starting fluentd-1.19.2 pid=35792 ruby="3.4.9"
2026-03-20 23:37:03 -0300 [info]: spawn command to main:  cmdline=["C:/Ruby34-x64/bin/ruby.exe", "-Eascii-8bit:ascii-8bit", "C:/Ruby34-x64/bin/flue
ntd", "-c", "fluent.conf", "--under-supervisor"]
2026-03-20 23:37:04 -0300 [info]: #0 init worker0 logger path=nil rotate_age=nil rotate_size=nil
2026-03-20 23:37:04 -0300 [info]: adding match pattern="app.logs" type="elasticsearch"
2026-03-20 23:37:04 -0300 [info]: adding source type="tail"
2026-03-20 23:37:04 -0300 [info]: #0 Oj is not installed, and failing back to JSON for json parser
2026-03-20 23:37:04 -0300 [info]: #0 starting fluentd worker pid=31840 ppid=35792 worker=0
2026-03-20 23:37:04 -0300 [info]: #0 following tail of ./app.log
2026-03-20 23:37:04 -0300 [info]: #0 fluentd worker is now running worker=0
```

Criação do Data View no Kibana para visualização dos logs enviados ao Elasticsearch.



Visualização dos logs da aplicação no Kibana através do Discover.